

BAYESIAN LIFE ESTIMATION WITH AN INVERSE GAUSSIAN PRIOR

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Summary: Bayesian analysis of the exponential model, based on life tests that are terminated after a preassigned number of failures, is carried out under the assumption of squared error loss function and an inverse Gaussian prior density on the parameter space. The Bayesian estimation of reliability for the case of 'attribute testing' has also been considered.

1. INTRODUCTION

Our starting point is the well-known exponential model of life-time governed by the probability density

$$(1.1) \quad f(x|\theta) = \frac{1}{\theta} e^{-x/\theta} \quad (0 < x < \infty; \theta > 0).$$

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We assume as in Epstein and Sobel (1953) that n items from a population specified by (1.1) are placed on a life test and the experiment is continued till the first r failures are observed, for some preassigned r . Let $x_1, x_2, \dots, x_r$ be the first r ordered observations in the random sample, $(n-r)$ being the number of survivals.

The object of the present paper is to derive the Bayesian estimators of mean life and reliability under the assumption of a squared error loss function and an inverse Gaussian prior distribution on the parameter space given by the pdf

$$(1.2) \quad g(\theta) = \left(\frac{\alpha}{2\pi}\right)^{\frac{1}{2}} \frac{1}{\theta^{3/2}} e^{-\frac{\alpha}{2m^2\theta}(\theta-m)^2} \quad (0 < \theta < \infty; \alpha, m > 0)$$

The case of 'attribute testing' is also considered and for details, the reader is referred to the paper of Bhattacharya (1967), wherein life testing and reliability analysis in a Bayesian framework were introduced for the first time.

2. BAYES ESTIMATOR OF MEAN LIFE

Under the assumptions stated earlier, the kernel of the likelihood of the sample is given by

$$(2.1) \quad L \propto \frac{1}{\theta^r} e^{-\frac{T_r}{\theta}}$$

where

$$(2.2) \quad T_r = \sum_{i=1}^r x_i + (n-r)x_r$$

is the accumulated life in the sample.

This likelihood is combined with the prior density (1.2) via Bayes' theorem to obtain the posterior density

$$(2.3) \quad g^*(\theta) \propto \frac{1}{\theta^{r+\frac{3}{2}}} e^{-\frac{\alpha}{2m^2} \cdot \theta - \frac{1}{\theta} [\frac{\alpha}{2} + T_r]} \quad (0 < \theta < \infty)$$

To obtain the normalization constant, we use the integral representation of the modified Bessel function of the third kind $K_\nu(Z)$ (cf. Erdélyi, 1953, p. 82, formula 23), viz.

$$(2.4) \quad \int_0^\infty u^{-\nu-1} e^{-\frac{Z}{2}(u + \frac{\mu^2}{u})} du = \frac{2}{\mu^\nu} K_\nu(\mu Z)$$

provided that $\operatorname{Re} Z > 0$ and $\operatorname{Re}(\mu^2 Z) > 0$. Use of this relation leads to the Bayesian posterior density

$$(2.5) \quad g^*(\theta) = C \cdot \frac{e^{-\frac{\alpha}{2m^2} \cdot \theta - \frac{1}{2\theta} [\alpha + 2T_r]}}{\theta^{r+\frac{3}{2}}} \quad (0 < \theta < \infty),$$

where

$$(2.6) \quad C = \frac{m^{r+\frac{1}{2}} [\alpha + 2T_r]^{\frac{1}{2}(r+\frac{1}{2})}}{2\alpha^{\frac{1}{2}(r+\frac{1}{2})} K_{r+\frac{1}{2}} \left(\sqrt{\frac{\alpha}{m^2}} [\alpha + 2T_r] \right)}$$

Under the assumption of squared error loss function, the Bayes estimator of mean life is simply the posterior mean and is, therefore, given by

$$(2.7) \quad \begin{aligned} \hat{\theta} &= C \int_0^\infty \frac{e^{-\frac{\alpha}{2m^2} \cdot \theta - \frac{1}{2\theta} [\alpha + 2T_r]}}{\theta^{r+\frac{1}{2}}} d\theta \\ &= C \cdot \frac{2\alpha^{\frac{1}{2}(r-\frac{1}{2})}}{m^{r-\frac{1}{2}} [\alpha + 2T_r]^{\frac{1}{2}(r-\frac{1}{2})}} K_{r-\frac{1}{2}} \left(\sqrt{\frac{\alpha}{m^2}} [\alpha + 2T_r] \right), \end{aligned}$$

on using (2.4). We substitute for C from (2.6), and obtain for the Bayes estimator

$$(2.8) \quad \hat{\theta} = \frac{m^2}{\alpha} \frac{U_r K_{r-\frac{1}{2}}(U_r)}{K_{r+\frac{1}{2}}(U_r)},$$

where

$$(2.9) \quad U_r = \frac{\alpha}{m} \left[1 + \frac{2T_r}{\alpha} \right]^{\frac{1}{2}}.$$

Since (2.8) involves a ratio of spherical Bessel functions, further simplification is possible by a formula of Erdélyi (op. cit., p. 10, formula 41), viz.

$$(2.10) \quad K_{n+\frac{1}{2}}(Z) = \left(\frac{\pi}{2Z} \right)^{\frac{1}{2}} e^{-Z} \sum_{k=0}^n (n+\frac{1}{2}, k) (2Z)^{-k}$$

where (ν, k) is Hankel's symbol defined by

$$(2.11) \quad (\nu, k) = \frac{\Gamma(\nu + \frac{1}{2} + k)}{k! \Gamma(\nu + \frac{1}{2} - k)}.$$

Using (2.10) in (2.8), we obtain the Bayes estimator

$$(2.12) \quad \hat{\theta} = \frac{m^2}{\alpha} U_r \frac{\sum_{k=0}^{r-1} (r-\frac{1}{2}, k) (2U_r)^{-k}}{\sum_{k=0}^r (r+\frac{1}{2}, k) (2U_r)^{-k}}.$$

By using the integral representation (2.4), the posterior variance V is obtained as

$$(2.13) \quad V = \frac{m^4}{\alpha^2} U_r^2 \frac{\sum_{k=0}^{r-2} (r-\frac{3}{2}, k) (2U_r)^{-k}}{\sum_{k=0}^r (r+\frac{1}{2}, k) (2U_r)^{-k}} - (\hat{\theta})^2 \quad (r \geq 2)$$

3. BAYESIAN RELIABILITY ANALYSIS

The reliability function of the exponential model is given by

$$R = R(t|\theta) = e^{-t/\theta} \quad (t > 0).$$

Under the assumption of the squared error loss function, the Bayes estimator of reliability is given by

$$(3.1) \quad \hat{R} = \int_0^{\infty} e^{-t/\theta} g^*(\theta) d\theta.$$

We substitute for $g^*(\theta)$ from (2.5) and use (2.4) to obtain the Bayes estimator

$$(3.2) \quad \hat{R} = \frac{U_r^{r+\frac{1}{2}}}{[V_r(t)]^{r+\frac{1}{2}}} \cdot \frac{K_{r+\frac{1}{2}}[V_r(t)]}{K_{r+\frac{1}{2}}[U_r]}.$$

where

$$(3.3) \quad V_r(t) = \frac{\alpha}{m} \left[1 + \frac{2T_r + 2t}{\alpha} \right]^{\frac{1}{2}}.$$

The variance of the posterior distribution of reliability is given by

$$(3.4) \quad V^* = \frac{U_r^{r+\frac{1}{2}}}{[W_r(t)]^{r+\frac{1}{2}}} \cdot \frac{K_{r+\frac{1}{2}}[W_r(t)]}{K_{r+\frac{1}{2}}[U_r]} - (\hat{R})^2,$$

where

$$(3.5) \quad W_r(t) = \frac{\alpha}{m} \left[1 + \frac{2T_r + 4t}{\alpha} \right]^{\frac{1}{2}}.$$

We now consider the case of 'attribute life testing' (cf. Ehrenfeld,

1962 and Bhattacharya, 1962). It is assumed here that n items from a population specified by (1.1) are placed on a life test and that after a fixed duration of time T , the number of survivals S is recorded. The exact failure times are not available even for the failed items. The likelihood of S survivals is

$$(3.6) \quad l_1 = \binom{n}{S} e^{-\frac{ST}{\theta}} \left(1 - e^{-\frac{T}{\theta}}\right)^{n-S} \quad (S = 0, 1, 2, \dots, n).$$

This is combined with the prior (1.2) to obtain the Bayesian posterior density

$$(3.7) \quad g^{**}(\theta) = C_0 \cdot \frac{e^{-\frac{ST}{\theta}} \left(1 - e^{-\frac{T}{\theta}}\right)^{n-S} e^{-\frac{\alpha}{2m^2} \cdot \theta - \frac{\alpha}{2\theta}}}{\theta^{3/2}} \quad (0 < \theta < \infty),$$

where C_0 is a normalization constant. To obtain C_0 , we use the binomial expansion for $(1 - e^{-T/\theta})^{n-S}$ and (2.4). We obtain

$$(3.8) \quad C_0^{-1} = \sum_{j=0}^{n-S} \binom{n-S}{j} (-1)^j \frac{2\alpha^{1/4}}{m^{1/2} [\alpha + 2ST + 2jT]^{1/4}} K_{\frac{1}{2}} \left(\frac{\alpha}{m} \sqrt{1 + \frac{2(S+j)T}{\alpha}} \right).$$

Thus, for attribute testing, the Bayes estimator of reliability is given by

$$(3.9) \quad R^{**} = \int_0^\infty e^{-t/\theta} g^{**}(\theta) d\theta \\ = \frac{\sum_{j=0}^{n-S} \binom{n-S}{j} (-1)^j [W_2(j)]^{-1/2} K_{\frac{1}{2}}[W_2(j)]}{\sum_{j=0}^{n-S} \binom{n-S}{j} (-1)^j [W_1(j)]^{-1/2} K_{\frac{1}{2}}[W_1(j)]},$$

where

$$(3.10) \quad W_1(j) = \frac{\alpha}{m} \left[1 + \frac{2(S+j)T}{\alpha} \right]^{\frac{1}{2}}$$

and

$$(3.11) \quad W_2(j) = \frac{\alpha}{m} \left[1 + \frac{2(S+j)T + 2t}{\alpha} \right]^{\frac{1}{2}}.$$

Since $K_1(Z) = \left(\frac{\pi}{2Z}\right)^{\frac{1}{2}} e^{-Z}$, (3.9) simplifies to

$$(3.12) \quad R^{**} = \frac{\sum_{j=0}^{n-S} \binom{n-S}{j} (-1)^j [W_2(j)]^{-1} \exp[-W_2(j)]}{\sum_{j=0}^{n-S} \binom{n-S}{j} (-1)^j [W_1(j)]^{-1} \exp[-W_1(j)]}$$

for the Bayes estimator of reliability.

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